EduVerse AI · Core Engineering & Science Study Guide

Unit 1: Computer Networks · Transport Layer Protocols
TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are transport layer protocols with three key differe
1. Connection: TCP is connection-oriented using a 3-way handshake (SYN, SYN-ACK, ACK), whereas UDP is connectionle
2. Reliability: TCP guarantees delivery through acknowledgments, checksums, and retransmissions, while UDP provides be
3. Ordering and Overhead: TCP ensures in-order delivery of data packets with higher overhead; UDP sends independent da

Unit 2: Database Management Systems · Concurrency Control
When transactions T1 and T2 execute concurrently, if T1 reads data item X after T2 writes X without committing, a dirty rea
Strict Two-Phase Locking (Strict 2PL) prevents dirty reads and cascading aborts. Under Strict 2PL, a transaction must hold

Unit 3: Engineering Mathematics · Calculus & Differentiation
The power rule states that d/dx [x^n] = n * x^(n-1).
For the function f(x) = 3*x^3 - 5*x + 7:
The first derivative is f'(x) = 9*x^2 - 5.
Evaluating f'(x) at x = 2:
f'(2) = 9 * (2)^2 - 5 = 9 * 4 - 5 = 36 - 5 = 31.

Unit 4: Systems Programming · Python Asyncio Concurrency
To limit concurrency in Python asyncio applications, an asyncio.Semaphore is used.
Example implementation:

```
import asyncio

async def fetch_task(sem: asyncio.Semaphore, task_id: int):
    async with sem:
        await asyncio.sleep(0.5)
        return f'Task {task_id} done'

async def main():
    sem = asyncio.Semaphore(3)
    tasks = [fetch_task(sem, i) for i in range(10)]
    results = await asyncio.gather(*tasks)
```

Unit 5: Distributed Systems · 2PC Lock Wait Timeout
In distributed database transactions operating under the Two-Phase Commit (2PC) protocol, the maximum lock wait timeou